If an n x n matrix has exactly one eigenvalue, is the matrix diagonalizable? What is the eigenspace when the matrix is diagonalizable?

Answer:

For an n x n matrix with exactly one eigenvalue, the matrix is diagonalizable if and only if its single eigenspace has a dimension equal to n which means it has n linearly independent eigenvectors.

Key Points:

1. **Diagonalizability Condition**:

   If the matrix has a single eigenvalue with multiplicity n, it is diagonalizable only if the dimension of the eigenspace is also n. This means all the eigenvectors must span the entire space.

2. **Eigenspace**:

   When the matrix is diagonalizable and has only one eigenvalue with dimension of the eigenspace equal to n, the matrix is actually a scalar matrix A = rI where r is the eigenvalue, and I is the identity matrix.

- In this situation, the eigenspace Er is the entire space R^n. Every non-zero vector in the space is an eigenvector.

So, in summary, a matrix with exactly one eigenvalue is diagonalizable if and only if it is a scalar multiple of the identity matrix, and its eigenspace, in that case, spans the whole space.